

Sudhir Kumar Singh

📍 Pune | ✉ sudhirsingh3923@gmail.com | ☎ +91 9771913689 | in LinkedIn | 📄 Github | 🏆 Leetcode

Education

Savitribai Phule Pune University, (B.E) - Artificial Intelligence and Data Science July 2023 – July 2027

- GPA: 9.3/10
- **Coursework:** Object Oriented Programming, Discrete Mathematics, Data Structures and Algorithms, Operating Systems

Skills

Languages: C++, C, Java, Python, C#, JavaScript, SQL
Web/Backend: Node.js, Express.js, React.js, .NET, REST APIs, JWT, OAuth
Database: MongoDB, MySQL, PostgreSQL
Data Analysis: Excel, Power BI, Pandas, NumPy, Matplotlib, Seaborn
Artificial Intelligence: GenAI, Machine Learning, LangChain, RAG

Experience

Backend Web Development Intern, Prodigy InfoTech, Feb 2025 – March 2025

- Engineered scalable backend services using **Node.js, Express.js,** and **MongoDB**, processing over **10,000+ user request**.
- Optimized database queries, reducing data retrieval time by **40%** through indexing and query restructuring.
- Designed and implemented **secure RESTful APIs**, improving request handling efficiency by **30%**.
- Integrated **Nodemailer** for automated email notifications, reducing manual intervention by **90%**.

Projects

AI-Powered Prescription scanner & Generic Medicine Platform Demo MedGenix

- Developed a smart platform to **scan prescriptions using OCR**, with dashboards for users and chemists.
- Integrated **Groq chatbot, generic medicine search,** nearby medical locator, **multilingual support,** and **real-time price comparison from 1mg.**
- Tools Used: **MERN Stack, LLM, MongoDB, JWT, Cloudinary, Tailwind CSS, Map APIs.**

Pregnancy Wellness Platform Pregnancy-Wellness

- Built a **comprehensive maternal health platform** with features like **appointment scheduling, meal logging, community forums,** and **personalized prenatal/postnatal care.**
- Integrated an **AI-powered chatbot** to provide instant guidance on nutrition, exercises, and general pregnancy queries.
- Implemented **personalized meal and exercise recommendations,** dynamically adjusting based on pregnancy stage and user preferences.
- Optimized database queries, improving **data retrieval speed by 50%** for user health records.
- Tools Used: **MERN Stack (MongoDB, Express.js, React.js, Node.js), JWT, Tailwind CSS, AI-based chatbot.**

AI-Powered YouTube Video Summarizer YouTube Summarizer

- Developed a system to **summarize YouTube videos and answer questions** using Retrieval-Augmented Generation (RAG).
- Integrated **FastAPI backend, React frontend,** and a **browser extension** for seamless user experience.
- Deployed as a **cross-browser extension,** enabling users to generate summaries and Q&A directly on YouTube platform.
- Tools Used: **FastAPI, React.js, LangChain, FAISS, Groq LLMs, Gemini Embeddings, YouTube Transcript API.**

Certificates

Full Stack Web Development Certification Apna College

- Covered technologies like HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MongoDB.

C# Programming Certification Microsoft

- Learned C# fundamentals, object-oriented programming, and .NET framework.

Postman API Fundamentals Certification Postman

- Gained expertise in API testing, request handling, and automation using Postman.